Lecture 21

Gauge fields: connections, curvature and holonomy

Objectives. *A bundle has local frames, but differentiating their coefficients depends on the frame. A connection corrects this derivative and specifies parallel transport. We derive its change law and show how curvature measures transport around a small loop. A flat connection can still have nontrivial transport around a hole. The Aharonov–Bohm phase is this effect for a charged particle.*

By the end of this lecture you will be able to:

- Derive the connection and curvature laws under a change of frame.
- Compute parallel transport with the correct multiplication order.
- Relate a small counter-clockwise loop to its curvature.
- Compute the flat Aharonov–Bohm holonomy and its gauge-invariant phase.

21.1 Covariant derivatives and minimal coupling

Let $E \to M$ be a smooth rank-r complex vector bundle. Write $\Gamma(E)$ for its smooth sections and $\Omega^k(M;E)$ for its E-valued k-forms. In a frame, the latter are columns of ordinary k-forms. Recall Lecture 20's row frame e and coefficient column σ, with $s = e\sigma$. A *connection* is a complex-linear map

$$\nabla\colon \Gamma(E) \longrightarrow \Omega^1(M;E), \qquad \nabla(fs) = \mathrm{d}f \otimes s + f\nabla s \quad \text{for } f \in C^\infty(M,\mathbb{C}).$$

We first construct a *bump function*, a smooth function equal to one near a chosen point $p \in M$, with support inside a chosen neighbourhood U. Take a coordinate ball centred at p with $\overline{B_R} \subset U$ and $0 < r < R$. Define $\rho(t) = e^{-1/t}$ for $t > 0$ and $\rho(t) = 0$ for $t \le 0$. Every positive-side derivative is $e^{-1/t}$ times a polynomial in $1/t$, by induction. All tend to zero at zero, so ρ is smooth. In the ball's coordinates the desired function is

$$f(x) = \frac{\rho(R^2 - |x|^2)}{\rho(R^2 - |x|^2) + \rho(|x|^2 - r^2)}.$$

The denominator is positive. The function is one on B_r and zero outside B_R. Its support is compact inside the chart, so extension by zero is smooth on M.

The connection is local. If s vanishes near p, choose such an f supported where $s = 0$. The Leibniz rule applied to $fs = 0$ gives $(\nabla s)_p = 0$. A section defined on U can instead be multiplied by a bump supported in U, equal to one near p, and extended by zero. Locality makes its derivative at p independent of the extension. This defines the local operator with

the same Leibniz rule and commutes with restriction. In a local frame, define the matrix of one-forms A by $\nabla e = eA$. The Leibniz rule gives

$$\nabla s = eD\sigma, \qquad D\sigma = \mathrm{d}\sigma + A\sigma.$$

The matrix A is the local *gauge potential*. It is a matrix of one-forms, rather than a single global matrix on a nontrivial bundle. Lecture 16's tangent connection is the real version of this definition, with real fibres, real linearity and real-valued functions.

A *Hermitian metric* is a smoothly varying positive Hermitian inner product on each fibre, antilinear in its first argument. A connection is *unitary* if it differentiates this inner product by the product rule. A *unitary frame* is an orthonormal basis in every fibre of its domain. In such a frame that rule is equivalent to $A^\dagger = -A$. The covariant product rule differs from the ordinary derivative of $\sigma^\dagger\tau$ by the term $\sigma^\dagger(A^\dagger + A)\tau$. Here σ and τ are coefficient columns of two local sections. For a line bundle, A is therefore imaginary. For an atlas with transitions in a matrix group G, its *G-frames* are the right translates $e_i a$, with $a \in G$. Require A to take values in its Lie algebra in these frames. This is the setting of the transport statements below.

For charge $q \in \mathbb{R}$, the physical potential of Lecture 15 is the real one-form $A_{\mathrm{em}} = -\phi\mathrm{d}t + \mathbf{A}\cdot\mathrm{d}\mathbf{x}$. The connection convention and its component derivatives are

$$A_{\mathrm{conn}} = -\frac{\mathrm{i}q}{\hbar}A_{\mathrm{em}}, \qquad D_t = \partial_t + \frac{\mathrm{i}q}{\hbar}\phi, \qquad D_i = \partial_i - \frac{\mathrm{i}q}{\hbar}A_i.$$

Here $\hbar > 0$ remains explicit. Thus $-\mathrm{i}\hbar D_i = -\mathrm{i}\hbar\partial_i - qA_i$. Replacing ordinary derivatives by these derivatives is *minimal coupling*. The real electromagnetic potential and the imaginary connection potential are distinct. For a Hermitian matrix potential A_{phys} written with $D = \mathrm{d} - \mathrm{i}g_{\mathrm{YM}}A_{\mathrm{phys}}$, the corresponding matrix is $A_{\mathrm{conn}} = -\mathrm{i}g_{\mathrm{YM}}A_{\mathrm{phys}}$. The coupling g_{YM} is a scalar, never a gauge map.

21.2 Frame changes and gauge transformations

Let $h\colon U \to \mathrm{GL}(r,\mathbb{C})$ be a smooth change of frame, $e' = eh$. This is the map h, not Planck's constant. The same section has coefficients $\sigma' = h^{-1}\sigma$. Differentiating the new frame gives

$$\nabla(eh) = e(Ah + \mathrm{d}h) = ehA', \qquad A' = h^{-1}Ah + h^{-1}\mathrm{d}h. \tag{21.1}$$

The extra derivative compensates for the changing frame. The following diagram expresses the resulting covariance.

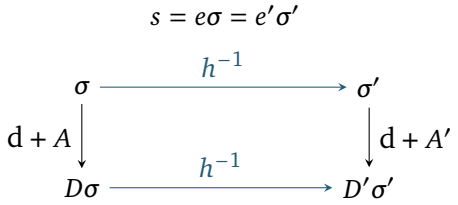


Figure 21.1: Both a section's coefficient column and its covariant derivative change by h^{-1}. The connection law makes the square commute.

Indeed, $\mathrm{d}(h^{-1}) = -h^{-1}(\mathrm{d}h)h^{-1}$ implies $D'\sigma' = h^{-1}D\sigma$ after the two derivative terms cancel. Conversely, local matrices obeying this change law on overlaps define a connection, because

these local derivatives agree as bundle-valued one-forms. This is Nakahara's compatibility law (10.9), translated to row frames.

The G-frames of E form a principal bundle by Lecture 20's cocycle construction. A *gauge transformation* is an equivariant smooth automorphism of this principal bundle over the identity of M. In local sections s_i, write it as $s_i(x)a \mapsto s_i(x)u_i(x)a$. The right-hand group coordinate is a. Since $s_j = s_i g_{ij}$, compatibility is precisely

$$u_j = g_{ij}^{-1} u_i g_{ij}.$$

A chosen global trivialization identifies these transformations with maps $M \to G$. For an abelian group the compatibility law reduces to $u_i = u_j$, even on a nontrivial bundle. For a nonabelian group, arbitrary global matrix maps need not satisfy it. A passive frame change describes the same connection. On E, the corresponding automorphism acts on columns by $\sigma_i \mapsto u_i \sigma_i$. Transporting the connection so that $D_i'(u_i\sigma_i) = u_i D_i \sigma_i$ gives

$$A_i' = u_i A_i u_i^{-1} + u_i \mathrm{d}(u_i^{-1}).$$

This follows by differentiating $u_i\sigma_i$ and cancelling $\mathrm{d}u_i$. It is the frame law with $h = u_i^{-1}$, the active description.

For electromagnetism take a real smooth function χ and $h = e^{-\mathrm{i}q\chi/\hbar}$. The frame law then reads

$$A'_{\mathrm{em}} = A_{\mathrm{em}} + \mathrm{d}\chi, \qquad \psi' = e^{\mathrm{i}q\chi/\hbar}\psi, \qquad D'\psi' = e^{\mathrm{i}q\chi/\hbar} D\psi.$$

Here ψ is the charged section's coefficient, which transforms inversely to its frame. These equations include the charge sign and agree with the minimal-coupling derivatives.

21.3 Curvature and the Bianchi identity

For matrix-valued forms, wedge multiplication includes matrix multiplication: $(A \wedge B)_{ab} = \sum_c A_{ac} \wedge B_{cb}$. Extend D to coefficient-valued forms by $D\eta = \mathrm{d}\eta + A \wedge \eta$. Its square on a coefficient column is multiplication by the *curvature two-form*

$$F = \mathrm{d}A + A \wedge A, \qquad F_{ij} = \partial_i A_j - \partial_j A_i + [A_i, A_j].$$

The components mean $F = \frac{1}{2} F_{ij} \mathrm{d}x^i \wedge \mathrm{d}x^j$. For an endomorphism-valued k-form B, the *covariant exterior derivative* is

$$DB = \mathrm{d}B + A \wedge B - (-1)^k B \wedge A.$$

This derivative acts on both the output and input of an endomorphism.

Theorem 21.1 (Local connection laws · *proved*). *A connection has local form $D = \mathrm{d} + A$ and the frame law (21.1). Its curvature transforms by $F' = h^{-1} F h$ and satisfies the Bianchi identity*

$$DF = \mathrm{d}F + A \wedge F - F \wedge A = 0.$$

Proof. The first assertions follow from the Leibniz and frame calculations above. Expanding $D^2\sigma$ cancels $-A \wedge \mathrm{d}\sigma$ against $A \wedge \mathrm{d}\sigma$, leaving $F\sigma$. To check curvature covariance explicitly, put $B = h^{-1}Ah$ and introduce the *Maurer–Cartan form* $\vartheta_{\mathrm{MC}} = h^{-1}\mathrm{d}h$. The product rule and the inverse derivative give

$$\mathrm{d}B = h^{-1}(\mathrm{d}A)h - \vartheta_{\mathrm{MC}} \wedge B - B \wedge \vartheta_{\mathrm{MC}}, \qquad \mathrm{d}\vartheta_{\mathrm{MC}} = -\vartheta_{\mathrm{MC}} \wedge \vartheta_{\mathrm{MC}}.$$

Since $A' = B + \vartheta_{\mathrm{MC}}$, substituting these identities into $\mathrm{d}A' + A' \wedge A'$ leaves only $h^{-1}(\mathrm{d}A + A \wedge A)h$. Finally,

$$\mathrm{d}F = \mathrm{d}A \wedge A - A \wedge \mathrm{d}A, \qquad A \wedge F - F \wedge A = A \wedge \mathrm{d}A - \mathrm{d}A \wedge A.$$

Adding gives the Bianchi identity. These are Nakahara's (10.38)–(10.47) in the present matrix notation. ∎

For U(1) the scalar one-form obeys $A \wedge A = 0$, so $F_{\mathrm{conn}} = -(\mathrm{i}q/\hbar)F_{\mathrm{em}}$ and the Bianchi identity is $\mathrm{d}F = 0$. The real source three-form $\mathcal{J}$ of Lecture 15 corresponds to $\mathcal{J}_{\mathrm{conn}} = -(\mathrm{i}q/\hbar)\mathcal{J}$ in connection units. The connection-valued source $\mathcal{J}_{\mathrm{conn}}$ is distinct from the real current. For a Hermitian matrix potential in the preceding convention, instead

$$F_{\mathrm{conn}} = -\mathrm{i}g_{\mathrm{YM}}(\mathrm{d}A_{\mathrm{phys}} - \mathrm{i}g_{\mathrm{YM}}A_{\mathrm{phys}} \wedge A_{\mathrm{phys}}).$$

The commutator term remains in the nonabelian theory. Within its matrix Lie algebra, the potential's value at a point can be changed arbitrarily by prescribing $\mathrm{d}h$ there with $h = I$. With $h = I$ there, curvature is unchanged. A general frame change conjugates it.

21.4 Parallel transport and a small square

A section along a piecewise smooth path $\gamma\colon [0,1] \to M$ is *parallel* if its covariant derivative along the path vanishes. In one local frame put $a(t) = A_{\gamma(t)}(\dot\gamma(t))$. Its coefficient equation is $\dot\sigma = -a(t)\sigma$. The *transport matrix* $U(t)$ sends the initial column to this column. Its initial value is I. Its equation is $\dot U = -a(t)U$. The coefficient equation gives the matrix equation column by column. Linear ODE existence and uniqueness give a solution throughout the compact parameter interval. The inverse solves $\dot V = Va(t)$, so $\frac{\mathrm{d}}{\mathrm{d}t}(VU) = 0$ and transport is invertible. On successive bundle charts, change coefficients at crossings and compose these solutions. The frame covariance proved above and uniqueness make this independent of subdivisions.

Iterating $U(t) = I - \int_0^t a(s)U(s)\mathrm{d}s$ gives

$$U(t) = I + \sum_{n \geq 1} (-1)^n \int_{0 < t_1 < \cdots < t_n < t} a(t_n) \cdots a(t_1)\mathrm{d}t_1 \cdots \mathrm{d}t_n.$$

The nth term has norm at most $K^n t^n / n!$ when $\|a\| \leq K$. The convergent series is the *path-ordered exponential*. Later times stand on the left. In a unitary frame for a unitary connection, $\frac{\mathrm{d}}{\mathrm{d}t}(U^\dagger U) = 0$, so transport is unitary. For scalar A this reduces to $U(\gamma) = \exp(-\int_\gamma A)$ in a frame covering the path. This is the U(1) transport in *Lie Groups*, Lecture 27, with the same sign.

The *holonomy* $\mathrm{Hol}(\gamma)$ is the transport around a based loop. Coefficient covariance $D'\sigma' = h^{-1}D\sigma$ and uniqueness show that changing the base frame conjugates it. Thus its trace, the *Wilson loop*, is invariant. A loop crossing several frames requires their transition matrices as well.

Theorem 21.2 (Parallel transport · *proved*). *Parallel transport is the unique solution*

$$\dot U(t) = -a(t)U(t), \qquad U(0) = I.$$

Under a change of frame it becomes

$$U'(\gamma) = h(\gamma(1))^{-1} U(\gamma) h(\gamma(0)).$$

For the counter-clockwise coordinate square of side ϵ based at 0,

$$\mathrm{Hol}(\square_\epsilon) = I - \epsilon^2 F_{xy}(0) + O(\epsilon^3).$$

Proof. The preceding construction proves existence, uniqueness and invertibility. The endpoint formula follows by differentiating $h(\gamma(t))^{-1}U(t)h(\gamma(0))$.

For the square, write $C = A_x(0)$, $D = A_y(0)$, $C_x = \partial_x A_x(0)$, $D_y = \partial_y A_y(0)$, $P = \partial_y A_x(0)$ and $Q = \partial_x A_y(0)$. Starting along positive x, integrate each edge equation through second order. The series bound makes all terms with at least three integrals $O(\epsilon^3)$:

$$\begin{aligned} U_1 &= I - \epsilon C + \frac{\epsilon^2}{2}(C^2 - C_x) + O(\epsilon^3),\\ U_2 &= I - \epsilon D + \epsilon^2(D^2/2 - Q - D_y/2) + O(\epsilon^3),\\ U_3 &= I + \epsilon C + \epsilon^2(C^2/2 + C_x/2 + P) + O(\epsilon^3),\\ U_4 &= I + \epsilon D + \frac{\epsilon^2}{2}(D^2 + D_y) + O(\epsilon^3). \end{aligned}$$

For example, on the second edge $a(t) = D + \epsilon Q + tD_y + O(\epsilon^2)$, for $0 \le t \le \epsilon$. The first integral contributes $-\epsilon D - \epsilon^2 Q - \epsilon^2 D_y/2$. The iterated integral contributes $\epsilon^2 D^2/2$. The other edges follow from their signed tangents in the same way. The successive transport acts on the left, giving

$$U_4U_3U_2U_1 = I + \epsilon^2(P - Q - [C, D]) + O(\epsilon^3) = I - \epsilon^2 F_{xy}(0) + O(\epsilon^3). \qquad \blacksquare$$

The figure fixes the traversal order used in this calculation.

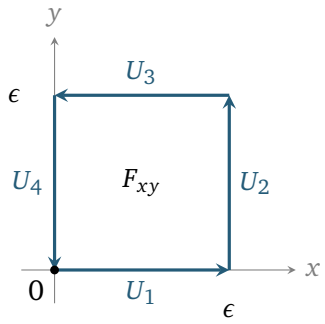


Figure 21.2: The path starts at 0 and traverses positive x, positive y, negative x, then negative y. Its transport is $U_4U_3U_2U_1$.

Recall that $\gamma \cdot \delta$ traverses γ first. Uniqueness therefore gives

$$U(\gamma \cdot \delta) = U(\delta)U(\gamma), \qquad U(\gamma^{-1}) = U(\gamma)^{-1}.$$

Transport reverses the order of the path product.

21.5 Flat connections and the Aharonov–Bohm phase

A connection is *flat* if $F = 0$. This condition removes local loop curvature but need not remove holonomy around a hole.

Theorem 21.3 (Flat holonomy · *sketched; proved for U(1)*)**.** *For a flat connection, based-loop holonomy depends only on the based homotopy class. After choosing a base G-frame, the map*

$$[\gamma] \longmapsto \mathrm{Hol}(\gamma)^{-1}$$

is a homomorphism $\pi_1(M, x_0) \to G$, where G is the matrix structure group. For each based loop its conjugacy class is independent of the chosen base frame.

Proof for U(1). In a small ball with a unitary frame, $\mathrm{d}A = 0$. By Theorem 13.2, write $A = \mathrm{d}f$, with imaginary f. For endpoints p_0 and p_1 in this ball, transport is $\exp(f(p_0) - f(p_1))$. Equivalently, Stokes gives unit transport around every loop bounding a disk in that ball. The endpoint formula also defines local transport on continuous paths. Subdividing into these balls extends it to continuous paths, agreeing with ODE transport on piecewise smooth paths. Refinements do not change it, since endpoint factors cancel. The local frames $e_i e^{-f_i}$ have zero connection form. On a connected overlap they therefore differ by a constant phase, by the frame law. Either local formula gives the same fibre map.

Given a continuous based homotopy, the Lebesgue lemma subdivides its parameter square into rectangles whose images lie in these balls. Choose a unit basis in the fibre at every grid vertex. Each rectangle has boundary product 1 by the local endpoint formula. Multiply the scalar boundary products. Every internal edge cancels with its reverse. The two outer side paths are constant at the base point and have transport 1. The bottom and top transports are therefore equal. Finally, taking inverses in the first-path-first composition rule gives the claimed homomorphism. This argument uses no global trivialization of the bundle. ∎

For a general matrix group, the variation of transport gives the corresponding sketch. Let $H(s,t)$ be a smooth based homotopy and $U(s,t)$ its transport from $t = 0$ to t. In a pullback frame fixed along both endpoint edges, the based boundary conditions give

$$U(s,1)^{-1}\partial_s U(s,1) = -\int_0^1 U(s,t)^{-1} F(H_s, H_t) U(s,t)\,\mathrm{d}t.$$

To obtain it, differentiate $\partial_t U = -A(H_t)U$ in s and combine $\partial_s U + A(H_s)U$ before integrating in t. Patching frames and the passage to continuous homotopies are omitted here. If $F = 0$, the formula makes the endpoint transport constant in s.

The angular one-form on the punctured plane is global although its angle is not:

$$\mathrm{d}\theta = \frac{-y\,\mathrm{d}x + x\,\mathrm{d}y}{x^2 + y^2}, \qquad M = \mathbb{R}^2 \setminus \{0\}.$$

The ideal thin solenoid has flux Φ positive out of the page. The two paths in the figure have the same endpoints.

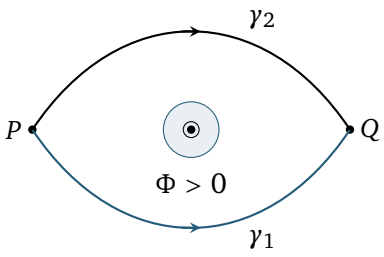


Figure 21.3: Both paths go from P to Q outside the solenoid. The loop $\gamma_1 \cdot \gamma_2^{-1}$ is counterclockwise and encloses positive flux.

Example 21.4 (The Aharonov–Bohm connection)**.** Compute the holonomy on the trivial complex line over M with

$$A_{\mathrm{em}} = \frac{\Phi}{2\pi}\mathrm{d}\theta, \qquad A = -\frac{\mathrm{i}q\Phi}{2\pi\hbar}\mathrm{d}\theta.$$

Differentiation gives $\mathrm{d}(\mathrm{d}\theta) = 0$ on M, so $F = 0$ there. Let γ be the counter-clockwise unit circle, with $\theta(t) = 2\pi t$ and circulation 2π. Hence

$$\mathrm{Hol}(\gamma) = \exp\left(-\oint_\gamma A\right) = e^{\mathrm{i}q\Phi/\hbar}.$$

The loop does not bound a disk in M, so Stokes does not set this circulation to zero. The bundle is trivial, but the connection need not be gauge equivalent to the zero connection. Put $\alpha = q\Phi/(2\pi\hbar)$. On the angular cover $h = e^{\mathrm{i}\alpha\theta}$ would cancel A. It descends to a single-valued map on M exactly when $\alpha \in \mathbb{Z}$. For an integer n, the genuine gauge map $h = e^{\mathrm{i}n\theta}$ changes α to $\alpha - n$ and leaves the holonomy unchanged. Thus these flat connections are distinguished by α modulo $\mathbb{Z}$.

The loop $\gamma_1 \cdot \gamma_2^{-1}$ winds once counter-clockwise, so its angular-form integral is also 2π by Lecture 3. Scalar transport gives $U(\gamma_1 \cdot \gamma_2^{-1}) = U(\gamma_1)U(\gamma_2)^{-1}$. In the ideal interference model, the ratio of the two transport phases is $U(\gamma_1)U(\gamma_2)^{-1} = e^{\mathrm{i}q\Phi/\hbar}$. For amplitudes c_1 and c_2 this changes the interference term to $2\,\mathrm{Re}(c_1\bar{c}_2 e^{\mathrm{i}q\Phi/\hbar})$. The phase is defined modulo 2π. The signed particle charge is q. This is the phase recalled in *Quantum Mechanics II*, Lecture 3, the physics box at the end of §3.4, *The momenta P_θ*, when forward hops run counter-clockwise. Nakahara, §10.5.3, (10.95)–(10.100), gives the physical minimal-coupling and interference calculation with $\hbar = 1$. His normalized imaginary-potential notation is translated here by the explicit charge ledger, rather than identifying his $\mathrm{i}A$ directly with our A_{conn}.

21.6 A constant nonabelian connection

The coefficient column σ is distinct from the Pauli matrices σ_x, σ_y and σ_z, recalled explicitly as

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Example 21.5 (Curvature from noncommuting transport)**.** Find the curvature and small-square holonomy of the connection on $\mathbb{R}^2 \times \mathbb{C}^2$ given, for $a, b \in \mathbb{R}$, by

$$A = C\mathrm{d}x + D\mathrm{d}y, \qquad C = -\frac{\mathrm{i}a}{2}\sigma_z, \qquad D = -\frac{\mathrm{i}b}{2}\sigma_x.$$

Both coefficient matrices are traceless and anti-Hermitian, so their transport lies in $\mathrm{SU}(2)$. Indeed, unitarity follows as above and $\frac{\mathrm{d}}{\mathrm{d}t}\det U = -\operatorname{tr}(A(\dot\gamma))\det U = 0$. The exterior derivative is zero, but $[\sigma_z, \sigma_x] = 2\mathrm{i}\sigma_y$ gives

$$F = [C, D]\mathrm{d}x \wedge \mathrm{d}y = -\frac{\mathrm{i}ab}{2}\sigma_y \mathrm{d}x \wedge \mathrm{d}y.$$

It is nonzero when $ab \neq 0$. The constant edge equations have exact solutions, so

$$\mathrm{Hol}(\square_\epsilon) = e^{\epsilon D}e^{\epsilon C}e^{-\epsilon D}e^{-\epsilon C} = I + \frac{\mathrm{i}ab\epsilon^2}{2}\sigma_y + O(\epsilon^3).$$

The coefficient is $-F_{xy}$, as the transport theorem requires. Thus vanishing ordinary derivatives of a nonabelian potential do not imply flatness.

Self-check. Show that a pure gauge $A = h^{-1}\mathrm{d}h$ is flat and compute its transport.

Answer. The inverse derivative gives $\mathrm{d}(h^{-1}\mathrm{d}h) = -h^{-1}\mathrm{d}h \wedge h^{-1}\mathrm{d}h$, which cancels $A \wedge A$. This is the Maurer–Cartan identity for $\vartheta_{\mathrm{MC}} = h^{-1}\mathrm{d}h$. Differentiating $U(t) = h(\gamma(t))^{-1}h(\gamma(0))$

verifies the transport equation and its initial condition. If h is global, every closed loop has holonomy I. The Aharonov–Bohm cancellation uses only a local h unless $\alpha \in \mathbb{Z}$.

Summary. A connection differentiates sections independently of their chosen frame. Its matrices change by $A' = h^{-1}Ah + h^{-1}\mathrm{d}h$. Curvature is $F = \mathrm{d}A + A \wedge A$, with $F' = h^{-1}Fh$ and $DF = 0$. Transport solves $\dot{U} = -A(\dot{\gamma})U$ and multiplies later segments on the left. A counter-clockwise small square has $\mathrm{Hol} = I - \epsilon^2 F_{xy} + O(\epsilon^3)$. A flat connection can retain nontrivial holonomy around a hole, as the Aharonov–Bohm phase $e^{\mathrm{i}q\Phi/\hbar}$ shows. Lecture 22 relates curvature flux to the transition winding of a line bundle.

Reading. Nakahara, §10.1.3–4, §10.3.4–5, §10.4.2 and §10.5.1, 10.5.3.